Rather than alert → It should simulate buy, sell actions of the corresponding stock. Update the document

Below is the updated **Functional Specification Document** for your QQQ leveraged ETF rules engine. It now simulates actual buy/sell actions of TQQQ, SQQQ, or cash, tracking portfolio allocation and running value—not just producing "alerts."

Functional Specification Document

QQQ Leveraged ETF Automated Trading Simulator

1. Overview

This program reads QQQ historical price data from Excel, calculates daily technical indicators, applies a defined rule book, and **simulates systematic buys/sells of TQQQ , SQQQ , and cash holdings**. It produces a daily trading ledger reflecting positions, transactions, and portfolio value over time.

2. Inputs

a) Data Source

- Excel file: "QQQ.xlsx" (provided by user)
- Required columns: Date, QQQ OHLC, Volume, TQQQ Price, SQQQ Price

b) Trading Parameters

- Initial portfolio balance (e.g., $10,000 or user-defined)
- SMA lookback periods: 50-day, 200-day
- ATR averaging window: 14 days
- **No leverage/borrowing beyond TQQQ/SQQQ exposures**

3. Technical Calculations

- Daily QQQ close
- 50-day and 200-day simple moving averages of QQQ close
- 14-day ATR (true range calculation)
- Detect trend of 50-day SMA (rising/flat/falling)

4. Simulation Logic (Fully Automated Trade Handling)

For each trading day:

- Calculate today's technicals (QQQ close, SMAs, ATR).
- Evaluate position rule:
  1. **100% TQQQ:** QQQ > 50d SMA & QQQ > 200d SMA, 50d SMA rising.
  2. **50% TQQQ / 50% Cash:** QQQ > both SMAs, 50d SMA flat/falling.
  3. **100% Cash:** QQQ in between SMAs or technical signals mixed.
  4. **100% SQQQ:** QQQ < 50d SMA & QQQ < 200d SMA and both SMAs falling.
  5. **100% Cash:** QQQ < both SMAs, but 50d not falling (flat/rising).
  6. Volatility override: If ATR > threshold, hold Cash (optional).
- Simulate **portfolio reallocation only when signal changes**:
  - If currently holding TQQQ and model shifts to SQQQ, first sell all TQQQ (at current TQQQ price), then buy SQQQ.
  - Log all trades (date, asset, buy/sell, price, quantity).
  - Update cash and holdings each day per position, compounding returns.
  - Record transaction costs (if desired; default to $0).
  - Portfolio value = (TQQQ * shares + SQQQ * shares + cash).

5. Ledger Book & Transaction Book

a) Daily Position/Trade Ledger (CSV/DataFrame)

- Date
- QQQ close
- 50d SMA, 200d SMA, ATR
- Signal
- Action taken (Buy/Sell/Hold, TQQQ/SQQQ/Cash, qty, price)
- Shares held (TQQQ, SQQQ, cash) after action
- Daily and cumulative portfolio value

b) Transaction Ledger (CSV/DataFrame)

- Trade date
- Asset (TQQQ, SQQQ)
- Buy/Sell
- Quantity
- Price
- Amount invested or cashed out
- Portfolio value before/after

6. Additional Requirements

- Simulated trades occur **end-of-day** at that day's closing price.
- No partial-day signals; stick to one action per calendar date.
- No leverage outside TQQQ/SQQQ instruments.
- Skip calculations if SMA/ATR periods not yet satisfied.

7. Outputs

- **Daily Position Ledger**: Step-by-step portfolio evolution and strategy signals
- **Transaction (Trade) Ledger**: All trades with buy/sell actions, sizes, and realized P/L
- **Summary Performance Report**: Final value, CAGR, drawdowns, trade stats (optionally)

8. Error Handling

- Handle missing price days (skip or fill forward as needed).
- Skip trading when any closing price is not available for a required symbol.

9. Implementation Notes

- Use Pandas for all data processing.
- Read Excel with openpyxl (or pandas.read_excel)
- Suggested structure:
  - main.py (program entry)
  - trade_engine.py (core rule logic, position tracking, indicators)
  - ledger.py (ledger io and formatting)
  - config.py (parameters and constants)

Example Position Ledger (excerpt):

Date	QQQ	50d SMA	200d SMA	ATR	Signal	Action	TQQQ sh	SQQQ sh	Cash	Value
2025-11-10	618.92	615	570	18.4	100% TQQQ	Buy TQQQ, All-in	176.20	0	$0	$10,000
2025-11-20	585.67	607.15	543.5	21.2	100% Cash	Sell TQQQ, All	0	0	$8,500	$8,500

This framework ensures any developer can implement an end-to-end, backtest-ready simulation of your trading strategy including all executions, asset flows, and portfolio value logic. Let me know if you want a boilerplate code template to jumpstart the build.

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1. https://www.perplexity.ai/finance/QQQ
2. https://www.perplexity.ai/finance/TQQQ
3. https://www.perplexity.ai/finance/SQQQ